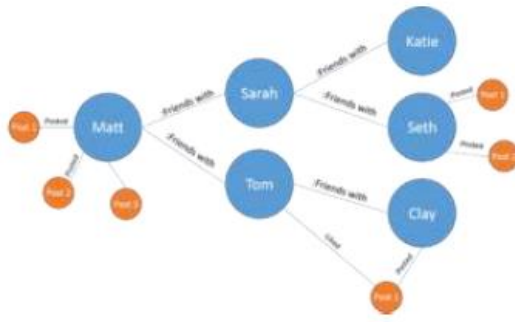


Q.1

18. Using the following database write the queries with Neo4j Cypher to:



a. Create above database

Ans. CREATE

```
(Matt:Person {name: "Matt"}),
(Sarah:Person {name: "Sarah"}),
(Tom:Person {name: "Tom"}),
(Katie:Person {name: "Katie"}),
(Seth:Person {name: "Seth"}),
(Clay:Person {name: "Clay"}),
(P_1:Post {title: "Post 1", author: "Matt"}),
(P_2:Post {title: "Post 2", author: "Matt"}),
(P_3:Post {title: "Post 3", author: "Matt"}),
(Post1_Seth:Post {title: "Post 1", author: "Seth"}),
(Post2_Seth:Post {title: "Post 2", author: "Seth"}),
(Post1_Clay:Post {title: "Post 1", author: "Clay"}),
```

```
(Matt)-[:FRIENDS_WITH]->(Sarah),
(Matt)-[:FRIENDS_WITH]->(Tom),
(Sarah)-[:FRIENDS_WITH]->(Seth),
(Sarah)-[:FRIENDS_WITH]->(Katie),
```

(Tom)-[:FRIENDS_WITH]->(Clay),

(Matt)-[:POSTED]->(P_1),

(Matt)-[:POSTED]->(P_2),

(Matt)-[:POSTED]->(P_3),

(Seth)-[:POSTED]->(Post1_Seth),

(Seth)-[:POSTED]->(Post2_Seth),

(Clay)-[:POSTED]->(Post1_Clay),

(Tom)-[:LIKED]->(Post1_Clay);

b.find Matts friends

ans.

```
1 match (Matt:Person {name:"Matt"})-[:FRIENDS_WITH]->(friends:Person)
2 return friends.name
```

	friends.name
1	"Sarah"
2	"Tom"

b. Find posts created by Matt

```
1 match (Matt:Person {name:"Matt"})-[:POSTED]->(p:Post)
2 return p.title
```

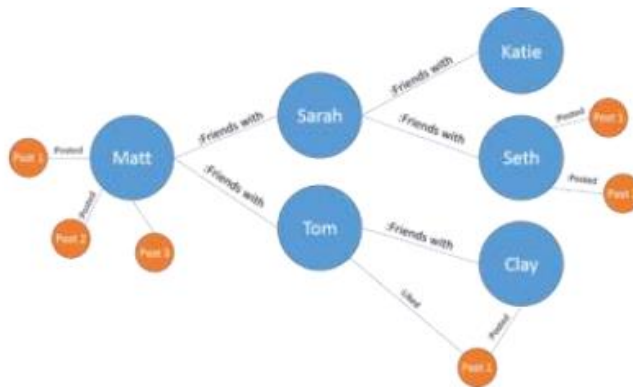
	p.title
1	"Post 1"
2	"Post 2"
3	"Post 3"

C.Find who like Post 1

```
1 match (p:Person)-[:LIKED]→(post:Post {title:"Post 1"})
2 return p.name
```

	p.name
1	"Tom"

Q.20



- create the above database.
- know that if Katie create any posts?
- Know how many friends does Seth have?
- Know that is there a path of friendship between Matt and Clay

a. ans

CREATE

```
(Matt:Person {name: "Matt"}),
(Sarah:Person {name: "Sarah"}),
(Tom:Person {name: "Tom"}),
(Katie:Person {name: "Katie"}),
(Seth:Person {name: "Seth"}),
(Clay:Person {name: "Clay"}),
(P_1:Post {title: "Post 1", author: "Matt"}),
```

```
(P_2:Post {title: "Post 2", author: "Matt"}),  
(P_3:Post {title: "Post 3", author: "Matt"}),  
(Post1_Seth:Post {title: "Post 1", author: "Seth"}),  
(Post2_Seth:Post {title: "Post 2", author: "Seth"}),  
(Post1_Clay:Post {title: "Post 1", author: "Clay"}),
```

```
(Matt)-[:FRIENDS_WITH]->(Sarah),  
(Matt)-[:FRIENDS_WITH]->(Tom),  
(Sarah)-[:FRIENDS_WITH]->(Seth),  
(Sarah)-[:FRIENDS_WITH]->(Katie),  
(Tom)-[:FRIENDS_WITH]->(Clay),
```

```
(Matt)-[:POSTED]->(P_1),  
(Matt)-[:POSTED]->(P_2),  
(Matt)-[:POSTED]->(P_3),  
(Seth)-[:POSTED]->(Post1_Seth),  
(Seth)-[:POSTED]->(Post2_Seth),  
(Clay)-[:POSTED]->(Post1_Clay),
```

```
(Tom)-[:LIKED]->(Post1_Clay);
```

b.ans

neo4j

The screenshot shows the Neo4j Cypher query editor interface. The query editor contains the following Cypher query:

```
1 match (Katie:Person {name : "Katie"})-[:POSTED]-(post)  
2 return post.title
```

Below the query editor, there is a sidebar with two tabs: "Table" and "Code". The "Table" tab is selected, and it displays the message "(no changes, no records)".

c.ans

```
1 match (Seth:Person {name : "Seth"})-[:FRIENDS_WITH]→(friends)
2 return count(friends)
```

	count(friends)
1	0

d.ans

neo4j\$

```
1 match path= shortestPath((m:Person {name:"Matt"})-
  [:FRIENDS_WITH]→(c:Person {name:"Clay"}))
2 return path;
3
```

	(no changes, no records)
--	--------------------------